PRODUCT REQUIREMENTS · NOVEMBER 2026

PRD: "Ask Your Data" trust layer for Snowsight

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1. PROBLEM

Business users (RevOps, finance, CS leaders) have questions that live in the warehouse but depend on an analyst to write SQL. Natural-language-to-SQL demos look magical, but adoption stalls on one failure: **a confident wrong answer**. One wrong number in a QBR and users stop asking. Accuracy alone is the wrong headline; the product question is *when should the assistant answer, and when should it say no?*

2. USERS AND JOBS

User	Job to be done	Pain today
Business analyst / RevOps	"Give me ARR by region this quarter without waiting on data eng"	Queue for analysts, inconsistent metric definitions
Data admin / governance owner	"Let people self-serve without exposing PII or inventing metrics"	Fear of ungoverned AI access
Data engineer	"Stop being the human SQL API"	Ad hoc requests

3. PRINCIPLES (WHAT THE PROTOTYPE PROVES)

1. **Governed definitions beat model cleverness.** "Revenue" resolves through a semantic model (MRR on active subscriptions) and the assumption is shown to the user.
2. **Abstain on purpose.** PII, writes, causal questions, and undefined metrics get a helpful refusal, not a guess.
3. **Policy lives below the model.** Read-only, table allow-list, and column masking are enforced in the database authorizer, so any engine (rules, LLM, future) inherits them.
4. **Measure trust, not just accuracy.** The scorecard tracks silent-error rate and over-refusal next to execution accuracy.

4. SUCCESS METRICS

Metric	Definition	Prototype result (38 questions)
Silent-error rate (north star, lower is better)	confident wrong answers / all questions	baseline 79% → semantic 5%
Execution accuracy	result-set match vs gold SQL, answerable questions	17% → 97%
Refusal recall	unsafe/unanswerable questions correctly declined	8 of 9 for the semantic engine (baseline: 2 of 9)
Over-refusal	answerable questions wrongly declined (cost of safety)	0%
Latency	p50 end-to-end	sub-millisecond for rules; LLM engine measured the same way

Production leading indicators: weekly active askers, % answers expanded to view SQL, thumbs-down rate, % follow-up clarifications, analyst-ticket deflection.

5. SCOPE

v1 (shipped here): metric/dimension/filter/time/ranking planner, assumptions + confidence, abstention, governance guard, eval harness, HTML scorecard. **Next:** (a) LLM engine behind the same guard (stub included; compare in the scorecard), (b) clarifying questions instead of flat refusals, (c) semantic-model authoring UX for admins, (d) ratio/derived metrics and period-over-period (the two residual failures below), (e) feedback loop turning thumbs-down into new eval cases.

6. KNOWN FAILURES (KEPT VISIBLE ON PURPOSE)

- *"Credits per dollar of MRR by segment"*: engine picks one metric and silently drops the other. Needs derived-metric support or a multi-metric detector that abstains.
- *"Which Enterprise customers have declining credit usage?"*: trend language is not recognised, so it answers a different question. Needs a "trend/decline" intent that abstains until supported.

7. RISKS

Risk	Mitigation
Eval set is small and author-written; the rules engine was tuned against it	Treat numbers as harness validation, not a benchmark claim. Next step: held-out set from real user questions
Semantic model drifts from business reality	Owner + review workflow for definitions; show assumptions in every answer
Over-refusal frustrates users	Track over-refusal as a first-class metric; add clarifying questions

8. ROLLOUT IDEA

Internal dogfood on one governed dataset → design partners (analytics-heavy accounts) → gated GA with the scorecard as the release criterion (ship only if silent-error rate < 2% on the customer's own eval set).